
Computer Graphics Report

Jiachen Lyu
5613244

University of Warwick

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1 Introduction

This project implemented a basic 3D FPS game using DirectX 12 and C++, featuring fundamental lighting, collision detection, materials, and animation systems. Instanced rendering was employed to support large-scale scenes and improve rendering efficiency. The models were provided in GEM format, and the stb image library was used for loading texture maps.

The project completed the setup of a standard rendering pipeline and implemented multiple vertex and pixel shaders to meet different rendering requirements. The project extends the base rendering framework provided in the course by introducing higher-level abstractions to improve code extensibility. In addition, a data-driven approach was adopted, with scene data loaded from CSV files to construct the game world.

2 Technical Implementation

2.1 Mesh Rendering

Considering that the same mesh may be reused multiple times within the project, a MeshManager class was implemented to manage mesh resources. Each mesh is loaded only once and reused when required, thereby reducing unnecessary GPU memory usage.

```
1 void loadmesh(Core* core, vector<vector<
    ANIMATED_VERTEX>> vertices, vector<vector<
    unsigned int>> indices, string meshname) {
2     if (meshesmanager.find(meshname) !=
        meshesmanager.end()) {
3     }
4 }
5 else {
6     vector<GeneralMesh*> meshes;
7     for (int i = 0; i < vertices.size(); i++) {
8         GeneralMesh* mesh = new GeneralMesh();
9         mesh->init(core, vertices[i], indices[i]);
10        meshes.push_back(mesh);
11    }
```

```
12    meshesmanager.insert({ meshname, meshes });
13 }
14
15
16 }
```

Listing 1: Load Meshes function

During model loading, each model only stores its mesh data without immediately initializing the mesh objects. After loading is complete, the mesh data is passed to the MeshManager class together with an associated mesh name. If a mesh with the given name does not already exist, the mesh is initialised using the provided data and stored in the MeshManager. When required, meshes are retrieved from the manager by name and used for draw calls.

2.1.1 Data-Driven

The project uses CSV files to store map data, including the distribution of all static and dynamic models in the scene. Three separate CSV files are used for this purpose. The first file stores information for general game objects, including their positions, the GEM models used, and the corresponding identifiers for the custom MeshManager and TextureManager. The second file stores data related to game objects rendered using instancing, while the third file records all position coordinates of instanced objects. After instancing was introduced into the project, the file for general game objects was primarily used to store data for animated objects only.

```
1 vector<Objectload> LoadData(vector<Objectload>*>
    scv1, vector<Objectload>*> scv2) {
2     ifstream file("../Resources/loadfile1.csv");
3
4     string line;
5
6     getline(file, line);
7     Objectload getinformation;
8     while (getline(file, line)) {
```

```

9      std::stringstream theline(line);
10     std::string word;
11
12     getline(theline, word, ',');
13     getinformation.type = word;
14     getline(theline, word, ',');
15     getinformation.name = word;
16     getline(theline, word, ',');
17     getinformation.position.x = stof(word);
18     getline(theline, word, ',');
19     getinformation.position.y = stof(word);
20     getline(theline, word, ',');
21     getinformation.position.z = stof(word);
22     getline(theline, word, ',');
23     getinformation.rotation.x = stof(word);
24     getline(theline, word, ',');
25     getinformation.rotation.y = stof(word);
26     getline(theline, word, ',');
27     getinformation.rotation.z = stof(word);
28     getline(theline, word, ',');
29     getinformation.location = (word);
30     getline(theline, word, ',');
31     getinformation.textureName = (word);
32     getline(theline, word, ',');
33     getinformation.iscollider = stoi(word);
34     getline(theline, word, ',');
35     getinformation.meshname = (word);
36     if (getinformation.type == "static") {
37         scv1->push_back(getinformation);
38     }
39     else {
40         scv2->push_back(getinformation);
41     }
42 }
43
44 file.close();
45 return scv;
46 }

```

Listing 2: Load CSV Method

2.1.2 Mesh Rendering and Alpha Testing

During mesh rendering, the vertices of a mesh are multiplied by the World matrix and the View-Projection matrix to transform them from model space to world space and then into clip space. The rendering pipeline subsequently performs clipping, after which the final fragments to be drawn are determined. During pixel rendering, alpha testing is applied to determine whether a pixel should be rendered based on its alpha value, thereby enabling the correct rendering of opaque meshes. After textures are introduced, the required pixel colour is obtained by sampling the texture according to the corresponding UV coordinates.

```

1 PS_INPUT VS(VS_INPUT input)
2 {
3     PS_INPUT output;
4     output.Pos = mul(input.Pos, W);
5     output.Pos = mul(output.Pos, VP);
6     output.Normal = mul(input.Normal, W);
7     output.Tangent = mul(input.Tangent, W);
8     output.TexCoords = input.TexCoords;
9     return output;
10 }
11 // The basic vertex transformation process

```

Listing 3: From Vertex to Pixel



Figure 1: Mesh Rendering and Alpha Testing

In the pixel shader stage, alpha testing is applied to models such as leaves and grass to discard transparent fragments. The figure above shows the result of mesh rendering, including opaque tree trunks, the T-Rex model, and tree leaves that have been discarded through alpha testing.

2.1.3 Vertex Animation

In the vertex shader, vertex offsets are applied to simulate grass swaying. First, an offset parameter is computed based on time and wind strength. A UV-based weighting factor is then calculated, where vertices closer to the root receive smaller offsets and those closer to the tip receive larger offsets. Periodic oscillations are applied to the x and z axes, and a spatial phase offset is introduced to prevent synchronized motion across all grass instances. The final displacement is obtained by scaling the offset by the UV weight and wind intensity.

```

1 float3 GetWindEffect(float3 VertexWorldPos,
2     float GrassUV, float WindSpeed, float
3     WindStreth, float GameTime)
4 {
5     float Timer = GameTime * WindSpeed;
6     float3 GrassWindOffset = float3(0, 0, 0);
7
8     float UVChange = 1 - abs(GrassUV);
9     //Due to the issue with the texture, the UV
10    weights did not achieve a satisfactory
11    result.
12    GrassWindOffset.z = UVChange * sin(cos(
13        Timer + VertexWorldPos.z)) * WindStreth;
14    GrassWindOffset.x = UVChange * cos(sin(
15        Timer + VertexWorldPos.x)) * WindStreth;
16    //Introduce world coordinates for pseudo-
17    randomness
18    GrassWindOffset.y = 0;
19    return GrassWindOffset;
20 }

```

Listing 4: Wind implement

2.1.4 Instancing of Meshes

instancing is used to reduce CPU workload and improve GPU utilization when rendering a large number of similar objects. An instance buffer is created on the GPU

to store per-instance transformation matrices, allowing multiple instances of the same mesh to be rendered in a single draw call. Since all instances share the same rendering pipeline and constant buffers, this approach is only suitable for objects with minimal variation.



Figure 2: Instancing with Grass

2.2 Basic Camera Control System

The camera control system consists of two main components: camera orientation control and camera movement control. Camera orientation is controlled using mouse input, while camera movement is handled through the keyboard using the W, A, S, and D keys.

Each frame, the mouse displacement along the x and y axes is captured and used to update the camera's yaw and pitch angles. Based on these values, the camera's forward direction vector is computed using the corresponding mathematical formulas, as illustrated in the code. Once the forward direction is obtained, it is added to the camera position to calculate the target "to" vector. The right direction is then derived using the cross product between the forward vector and the world up direction. Finally, a LookAt matrix is constructed to update the camera orientation.

During rotation, limits are applied to the pitch angle to prevent the camera from flipping.

```
1 forward.x = cosf(pitch) * cosf(yaw);
2 forward.y = sinf(pitch);
3 forward.z = cosf(pitch) * sinf(yaw);
```

Listing 5: Convert the Angle to Forward Vector

Camera movement is performed relative to the current viewing direction, with the W, S, A, and D keys. Each frame, keyboard input is processed to determine the movement direction vector based on the camera's forward and right vectors. This direction vector is multiplied by the movement speed and the frame time to obtain the final displacement, which is then added to the current camera position to produce the updated position. Since the camera is constrained to move on the ground plane and does not move along the y-axis, the x and z components of the forward and right vectors are normalized independently before computing the movement. This prevents the movement speed from

decreasing as the camera's viewing direction deviates further from the horizontal plane.

```
1 Vec4 groundright = Vec4(right.x, 0, right.z, 0).
  normalize();
2 Vec4 groundforawrd = Vec4(forward.x, 0, forward.
  z, 0).normalize();
3 Vec4 delta(0, 0, 0, 0);
4 if (win.keys['A']) delta += Vec4(groundright.x,
  0, groundright.z, 0);
5 if (win.keys['D']) delta -= Vec4(groundright.x,
  0, groundright.z, 0);
6 if (win.keys['W']) delta += Vec4(groundforawrd.x,
  0, groundforawrd.z, 0);
7 if (win.keys['S']) delta -= Vec4(groundforawrd.x,
  0, groundforawrd.z, 0);
8 //delta=delta.normalize();
9 delta *= cameramovespeed * rexdt;
10 from = from + delta;
```

Listing 6: Move Control

2.3 Animation Control

The project uses skeletal animation. Data such as the position and rotation of each bone is stored, and smooth motion is achieved over time through interpolation. The resulting data is then uploaded to the shader, where the final animation calculations are performed.

This project uses two animated models, the T-Rex and the UZI, each with its own animation control logic. The animation system provided in the course supports animation switching and looping, and this project builds upon it to implement gameplay-driven animation control.

For the T-Rex, three animations are used: attack, run, and death. Each frame, the T-Rex's state is evaluated to select the appropriate animation. If its health drops below zero, the death animation is triggered and played until completion, after which the T-Rex is respawned. If the T-Rex is alive, a collision with the player triggers the attack animation; otherwise, the run animation is played. Damage is applied after each completed attack animation.

```
1 void anim(float rexdt, string status, Vec3 from)
  {
2   if (status == "death") {
3
4     if (status != Animatestatus) {
5       Animatestatus = status;
6       isdying = true;
7       dieanim = 0;
8     }
9   }
10  else {
11    if (enemymodelinstace.animationFinished
12    () == true || dieanim > 2.8999f) {
13      health = 100.f;
14      Animatestatus = "run";
15
16      die(from);
17      isdying = false;
18      dieanim = 0;
19    }
20    else {
```

```

21     enemymodelinstance.update("death",
22     rexdtd);
23     dieanim += rexdtd;
24 }
25
26 }
27
28 }
29 }
30 else {
31     Animatestatus = status;
32     enemymodelinstance.updatewithControl(
33     Animatestatus, rexdtd);
34     if (enemymodelinstance.animationFinished())
35     {
36         enemymodelinstance.resetAnimationTime();
37         attatcktime = 0;
38     }
39 }

```

Listing 7: TReX Animation Control Logic

The UZI control logic uses five animations and supports aiming, firing, and reloading. Reloading can be triggered manually by pressing the R key or automatically when the ammunition is depleted. Once the reload animation begins, it cannot be interrupted by any other animation.

Pressing the right mouse button switches the weapon into the aiming state, while pressing the left mouse button checks the ammunition count and firing cooldown before triggering the firing animation. The firing cooldown is defined by the duration of the firing animation, which loops continuously while the firing conditions are satisfied. Non-aimed firing is performed by pressing the left mouse button alone and follows logic similar to that of aimed firing.

```

1 void anim(Vec3 from, vector<Enemies*> enemies,
2   bool mouse[3], float rexdtd, Vec3 forawrd,
3   bool changeR, vector<Fire*> fires) {
4   if (changeR && !ischange) {
5       ischange = true;
6       heromodelinstance.update("17 reload", rexdtd);
7       //start reload
8   }
9   if (bullet <= 0 && !ischange) {
10      ischange = true;
11      heromodelinstance.update("17 reload", rexdtd);
12      //auto reload
13  }
14  if (ischange) {
15      heromodelinstance.update("17 reload", rexdtd);
16      if (heromodelinstance.animationFinished()) {
17          heromodelinstance.resetAnimationTime();
18          ischange = false;
19          bullet = 30;
20          shot = true;
21      }
22  }
23  else {
24      if (!shot) {
25          timecount += rexdtd;
26          if (timecount > cooldown) {
27              shot = true;
28              timecount = 0.f;
29          }
30      }
31  }
32  }

```

```

33  if (mouse[1]) {
34      //zoom
35      if (mouse[0] && bullet > 0) {
36          if (shot) {
37              // zoom shot
38              raytest.init(from+forward*3, forawrd);
39              //plus 3 to make bullet shot front
40              of camera
41              for (int i = 0; i < enemies.size(); i
42              ++){
43                  float t;
44                  if (enemies[i]->enemymodel.collision
45                  .rayAABB(raytest, t)) {
46                      //hit the target or not
47                      enemies[i]->health -= 5;
48                      for(int i=0;i< fires.size();i++){
49                          if(!fires[i]->active){
50                              fires[i]->work(rexdtd, from +
51                              (forawrd * (t + 3)));
52                              //particle effect
53                          }
54                      }
55                      break;
56                  }
57                  if (enemies[i]->health <= 0&&!
58                  enemies[i]->isdying) {
59                      score += 5;
60                      enemies[i]->isdying = true;
61                  }
62                  break;
63              }
64          }
65          bullet--;
66          shot = false;
67      }
68      heromodelinstance.update("13 zoom fire",
69      rexdtd);
70  }
71  if (heromodelinstance.animationFinished())
72  {
73      heromodelinstance.resetAnimationTime();
74      shot = true;
75  }
76  }
77  else {
78      heromodelinstance.updatewithControl("11
79      zoom idle", rexdtd);
80  }
81  }
82  else if (mouse[0] && bullet > 0) {
83      //normal shot
84      if (shot) {
85          raytest.init(from + forward * 3, forawrd
86          );
87          for (int i = 0; i < enemies.size(); i++)
88          {
89              float t;
90              if (enemies[i]->enemymodel.collision
91              .rayAABB(raytest, t)) {
92                  enemies[i]->health -= 5;
93                  for (int i = 0; i < fires.size(); i
94                  ++){
95                      if (!fires[i]->active) {
96                          fires[i]->work(rexdtd, from + (
97                          forawrd * (t+3)));
98                      }
99                  }
100              }
101          }
102      }
103      bullet--;
104      shot = false;
105  }
106  }

```



```

90
91
92         break;
93     }
94 }
95 if (enemies[i]->health <= 0 && !
enemies[i]->isdying) {
96     score += 5;
97     enemies[i]->isdying = true;
98 }
99     break;
100 }
101
102 }
103 bullet--;
104 shot = false;
105 }
106 heromodelinstance.update("08 fire", rexdt);
107 if (heromodelinstance.animationFinished())
{
108     heromodelinstance.resetAnimationTime();
109     shot = true;
110 }
111 }
112 else {
113     heromodelinstance.updatewithControl("04
idle", rexdt);
114     //idle
115 }
116 }
117 }

```

Listing 8: UZI Animation Control Logic

2.4 Collision System

This project employs both AABB collision detection and ray-based collision detection. Enemy T-Rex objects use a minimum bounding box as their collision box.

To generate the minimum bounding box, all vertices of the model are traversed to obtain the maximum and minimum values along the x, y, and z axes, forming an AABB in model space. This bounding box is then transformed into world space based on the object's position to obtain the final collision box.

Collision is determined by checking whether two bounding boxes overlap along each axis. If, in any direction, one bounding box lies completely on one side of the other, then they cannot intersect. If this does not occur on any axis, the two bounding boxes are considered to be colliding.

```

1 bool AABBtest(const Vec3& minB, const Vec3& maxB
2 )
3 {
4     if (realmaxpoint.x < minB.x || realminpoint.x
> maxB.x)
5         return false;
6     if (realmaxpoint.y < minB.y || realminpoint.y
> maxB.y)
7         return false;
8     if (realmaxpoint.z < minB.z || realminpoint.z
> maxB.z)
9         return false;
10
11     return true;
12 }

```

Listing 9: Collision Test

To handle rotation, the eight corner vertices of the bounding box are first constructed from the minimum and maximum points. All vertices are then rotated, and new minimum and maximum values are recalculated from the transformed vertices. The correct update order is to first compute the rotated minimum and maximum points in model space using the rotation matrix, and then add the object's position to obtain the final collision box in world space.

```

1 void update(Vec3 position, Matrix scv) {
2     //rotation need calculate the 8 corner point
and find the new min and max point
3     vector<Vec3>corner;
4     corner.push_back(Vec3(minpoint.x, minpoint.y,
minpoint.z));
5     corner.push_back(Vec3(minpoint.x, minpoint.y,
maxpoint.z));
6     corner.push_back(Vec3(minpoint.x, maxpoint.y,
minpoint.z));
7     corner.push_back(Vec3(minpoint.x, maxpoint.y,
maxpoint.z));
8     corner.push_back(Vec3(maxpoint.x, minpoint.y,
minpoint.z));
9     corner.push_back(Vec3(maxpoint.x, minpoint.y,
maxpoint.z));
10    corner.push_back(Vec3(maxpoint.x, maxpoint.y,
minpoint.z));
11    corner.push_back(Vec3(maxpoint.x, maxpoint.y,
maxpoint.z));
12    for (int i = 0; i < 8; i++) {
13        corner[i] = scv.MulVec3(scv, corner[i]);
14    }
15    realminpoint = Vec3(11111, 11111, 11111);
16    realmaxpoint = Vec3(-11111, -11111, -11111);
17    for (int i = 0; i < 8; i++) {
18        if (realmaxpoint.x < corner[i].x) {
19            realmaxpoint.x = corner[i].x;
20        }
21        if (realmaxpoint.y < corner[i].y) {
22            realmaxpoint.y = corner[i].y;
23        }
24        if (realmaxpoint.z < corner[i].z) {
25            realmaxpoint.z = corner[i].z;
26        }
27        if (realminpoint.x > corner[i].x) {
28            realminpoint.x = corner[i].x;
29        }
30        if (realminpoint.y > corner[i].y) {
31            realminpoint.y = corner[i].y;
32        }
33        if (realminpoint.z > corner[i].z) {
34            realminpoint.z = corner[i].z;
35        }
36    }
37    realminpoint += position;
38    realmaxpoint += position;
39
40
41 }

```

Listing 10: rotation collider box

The project uses ray casting to determine whether a shot hits an object. For ray-box intersection, the entry and exit times of the ray are computed separately along each axis. If an object lies behind the ray, the computed entry and exit times will be negative, in which case the entry and exit values need to be reversed; this is handled automatically using value comparisons.

The maximum entry time and the minimum exit time

are then selected as the actual entry and exit times of the ray. A collision is determined by comparing these times: if the exit time occurs before the entry time, no collision is detected. The variable t represents the time at which the ray enters the bounding box.

Objects located behind the ray are treated as non-colliding, as shooting in the project is directional and cannot hit objects located behind the firing direction

```

1 bool rayAABB(const Ray& r, float& t)
2 {
3     Vec3 s = (realminpoint - r.o) * r.invdir;
4     Vec3 l = (realmaxpoint - r.o) * r.invdir;
5     //calculate the time of input and output in
        xyz directions
6     Vec3 s1 = Min(s, l);
7     Vec3 l1 = Max(s, l);
8     //if the direction is negative, swap the value
9     float ts = max(s1.x, max(s1.y, s1.z));
10    //the max number in min,the real enter time
11    float tl = min(l1.x, min(l1.y, l1.z));
12    //the min number in max, the real leave time
13    t = min(ts, tl);
14
15
16    if (tl < 0.0f)
17        return false;
18    //direction is negative
19
20
21    return (ts < tl);
22
23
24 }
```

Listing 11: Ray Test

2.5 Normal Mapping, Lighting, and Textures

In the pixel shader, textures are sampled using UV coordinates to obtain the final pixel colour. Textures are stored in the TextureManager class using a map structure and are accessed by name. Textures for GEM models are loaded during the model loading stage, while textures for custom geometry are loaded in the main function and assigned after initialization.

```

1 void load(Core* core, vector<string> name,vector
        <string>NHname, string texturename)
2 {
3     if (textures.find(texturename) != textures.end
        ()) {
4
5     }
6     else {
7         vector <Texture*> scv;
8         vector <Texture*> scv1;
9         for (int i = 0; i < name.size(); i++) {
10
11
12             Texture* texture = new Texture;
13
14             texture->load(core, name[i]);
15             scv.push_back(texture);
16
17
18             Texture* texture1 = new Texture;
19
```

```

20         texture1->load(core, NHname[i]);
21         scv1.push_back(texture1);
22
23     }
24     textures.insert({ texturename, scv });
25     NHtextures.insert({ texturename, scv1 });
26 }
27
28
29 }
```

Listing 12: Load Textures Function

The project implements directional lighting. The light direction and strength are passed to the pixel shader through the pixel shader constant buffer. The light direction is first normalized and inverted to obtain the direction from the surface point to the light source. The dot product between the surface normal and the light direction is then computed to obtain the cosine of the angle between them. If this value is less than zero, the surface receives no light. The result is multiplied by the light strength to obtain the final brightness of the pixel.

A basic ambient lighting term was also introduced to ensure a minimum level of illumination.

```

1 float3 ComputeDirectionalLight(Light L, float3
        normal )
2 {
3     float3 lightVec = -normalize(L.Direction); //
        calculate the vector from surface to light
4     float ndotl = max(dot(lightVec, normal), 0.0
        f); //Lambert Cosine Law, calculate the
        strength of light
5     float3 lightStrength = L.Strength * ndotl; //
        mix
6
7     return lightStrength;
8 }
```

Listing 13: Light Function

To achieve more accurate lighting results, normal mapping is used in this project. After sampling the normal map using the UV coordinates, the sampled normal is transformed using the TBN basis to obtain a world-space normal, which is then used in the lighting calculations.

```

1 float4 PS(PS_INPUT input) : SV_Target0
2 {
3
4     float4 colour = tex.Sample(samplerLinear,
        input.TexCoords);
5     if (colour.a < 0.5f)
6         discard;
7     float3 normalTS = NHtex.Sample(samplerLinear
        , input.TexCoords).xyz;
8     normalTS = normalize(normalTS * 2.0f - 1.0f)
        ; // turn Vec to normal
9
10
11     float3 N = normalize(input.Normal);
12     float3 T = normalize(input.Tangent);
13     T = normalize(T - dot(T, N) * N);
14     float3 B = cross(N, T);
15     float3x3 TBN = float3x3(T, B, N);
16     float3 normalWS = normalize(mul(normalTS,
        TBN)); //turn to world space normal
17
```

```

18  Light L;
19  L.Strength = Strength;
20  L.Direction = normalize(Direction);
21
22  float4 ambient = gAmbientLight ;//ambient
  light
23
24  float4 directLight = ComputeLighting(L,
  input.PosW, normalWS);
25  float3 lighting = ambient.rgb + directLight.
  rgb;
26  float3 finalColor = colour.rgb * lighting;
27  return float4(finalColor, colour.a);
28 }

```

Listing 14: Pixel Shader for Lighting

```

1 PS_INPUT VS(VS_INPUT input)
2 {
3     PS_INPUT output;
4
5     output.Pos = float4((input.Pos.x-960)/960, (
  input.Pos.y-540)/540, 0.0f, 1.0f);
6     //calculate to [-1,1]
7
8     output.Normal = input.Normal;
9     output.Tangent = input.Tangent;
10
11     output.TexCoords = input.TexCoords;
12
13     return output;
14 }
15 }

```

Listing 15: Vertex Shader for UI

2.6 Other Effects

2.6.1 Skybox

In this project, the skybox effect is implemented using a custom spherical mesh combined with a skybox texture. During rendering, the skybox participates in depth testing but does not write to the depth buffer, ensuring that it always appears as a background element and does not occlude other objects in the scene.

The position of the skybox is updated to follow the camera position, creating the visual effect of an infinitely distant sky.



Figure 3: skybox

2.6.2 UI

The project implements several basic UI elements. These UI elements do not participate in depth testing and are rendered last, ensuring that they are never occluded by other objects in the scene. Numeric information is rendered using a bitmap font and is updated in real time. The bullet icon in the lower-left corner is updated dynamically based on the current remaining ammunition count.

The UI is essentially implemented using planes with textures. The vertex positions of these planes represent pixel positions on a 1920×1080 screen. During rendering, the vertex coordinates are transformed directly into screen space rather than undergoing the standard sequence of spatial transformations.

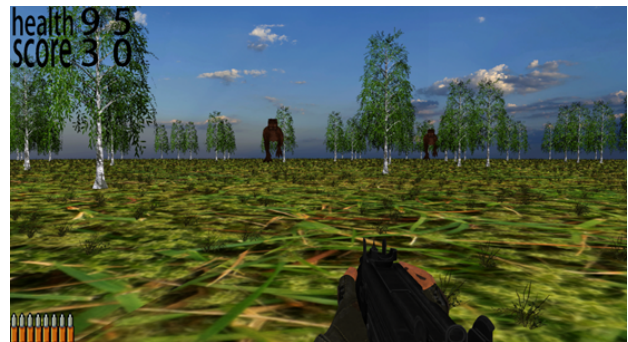


Figure 4: UI

2.6.3 Particles Effect

This project implements a spark-like particle effect. After a successful ray intersection is detected, a burst of spark particles is rendered at the position of the hit enemy.

The particle effect is rendered on a billboard, with the center of the plane representing the explosion origin. Pseudo-random vectors are generated in three directions and normalized to define the movement direction of each particle, while particle speeds are also determined using pseudo-random values. Each particle's position is updated over time, and in the pixel shader the distance between the particle and the current pixel is computed. The pixel brightness is then smoothly interpolated based on this distance.

Each particle effect consists of 109 particles. The shader iterates over all particles and accumulates their colour contributions to produce the final pixel colour.

To ensure consistent and stable particle motion within a single particle effect, pseudo-random values are generated based on position information; otherwise, the particle behaviour would appear completely chaotic.



Figure 5: Particle Effect

```

1 float3 FireworkParticles( float3 worldPos, float3
    fireworkPos, float time, float3 baseColor
2 )
3 {
4     float seed = hash1(dot(fireworkPos, float3
    (12.9898, 78.233, 45.164)));
5     //random seed
6     float3 color = float3(0, 0, 0);
7
8     const int NUM_PARTICLES = 109;
9
10    for (int i = 0; i < NUM_PARTICLES; i++)
11    {
12        float speed = lerp(3.0, 6.0, hash1(i *
    seed * 31.35));
13        //get random speed form 3 to 6
14        float3 dir = normalize(float3(
15            hash1(i*seed * 33.95)*2-1,
16            hash1(i*seed * 7.1)*2-1,
17            hash1(i*seed * 25.12)*2-1
18        ));
19        //random direction
20        float3 p = fireworkPos + dir * speed *
    time;
21        //now position of particle
22        float dist = length(worldPos - p);
23        float size = 0.1;
24        float fade = saturate(1.0 - time);
25        float intensity = smoothstep(size, 0.0,
    dist) * fade;
26        //calculate color based on distance
27        //the dis more close to zero more light
28        color += baseColor * intensity;
29    }
30    return color;
31 }

```

Listing 16: Particle Effect implement

3 Performance Analysis

This project uses tree models for performance testing. Performance is evaluated by comparing the frame rates achieved using the standard rendering approach and the instanced rendering approach under the same number of rendered objects. This comparison highlights the impact of CPU performance bottlenecks caused by multiple draw calls on overall frame rate.

In the test scene, directional lighting is applied to the trees. In addition, the scene also includes a skybox. The tests were conducted on a system equipped with an NVIDIA GeForce RTX 5070 Laptop GPU (8 GB) and an AMD Ryzen 9 8945HX with Radeon Graphics.



Figure 6: test scene

Table 1: Performance Test Results Under Different Conditions

Render Method	Tree Number	FPS
Normal	100	337
Normal	200	195
Normal	300	133
Normal	400	101
Normal	500	81
Normal	600	61
Normal	800	47
Normal	1000	39
Instancing	100	2747
Instancing	200	1653
Instancing	300	1389
Instancing	400	940
Instancing	500	745
Instancing	600	653
Instancing	800	514
Instancing	1000	412

The table clearly shows that when using the standard rendering approach, CPU performance becomes a significant bottleneck, severely limiting GPU utilization and resulting in a performance gap of nearly an order of magnitude. Without instancing, GPU utilization would become extremely low, making the application difficult to run smoothly.

4 Limitations

The overall structure of the project is relatively disorganized, as it evolved gradually from daily practice exercises without an initial architectural design. Although a structural refactor was carried out during the middle stage of development, the final structure resulted in a tightly nested dependency hierarchy, where classes depend on each other in multiple layers. As a

result, adding new features requires careful consideration of dependency relationships, which makes further development more difficult.

The UZI model exhibits noticeable texture jittering. This issue disappears when depth testing is disabled, indicating that it is likely caused by z-fighting. One possible cause is that multiple meshes within the model are positioned too closely to each other.

5 Future Work

Undoubtedly, the project needs a better overall structure, not only in the code. Since the project uses many external files, clear rules for file paths are needed. Random placement of shaders, assets, and CSV files increases the chance of errors.

The project uses separate classes for different object types. Many of these classes have similar functionality but are not clearly related. Adding a proper inheritance structure can reduce duplicated code and improve maintainability.

In the texture management system, textures are stored in groups using a map, which can lead to redundant storage when different groups share the same textures. A better approach would be to store each texture only once and reference them by index.

6 Conclusion

This project implemented an FPS game using C++ and DirectX 12. The game includes interaction logic, animation switching, textures, a skybox, particle effects, instancing, lighting with normal mapping, vertex animation, and other related features. Through the use of instancing, the project is able to render a large number of grass objects while maintaining smooth performance. Spark-like particle effects are used to simulate impact effects when a target is hit, and reasonable animation transition logic is implemented for both the player character and enemies.

Although there is still room for improvement, the project has a relatively high level of completion. During development, the project also increased my understanding of the rendering pipeline, which will be beneficial for future game development. Overall, the project achieved its intended objectives.

Project Repository

The source code for this project is available on GitHub:
<https://github.com/AODINGLVP/newworld.git>